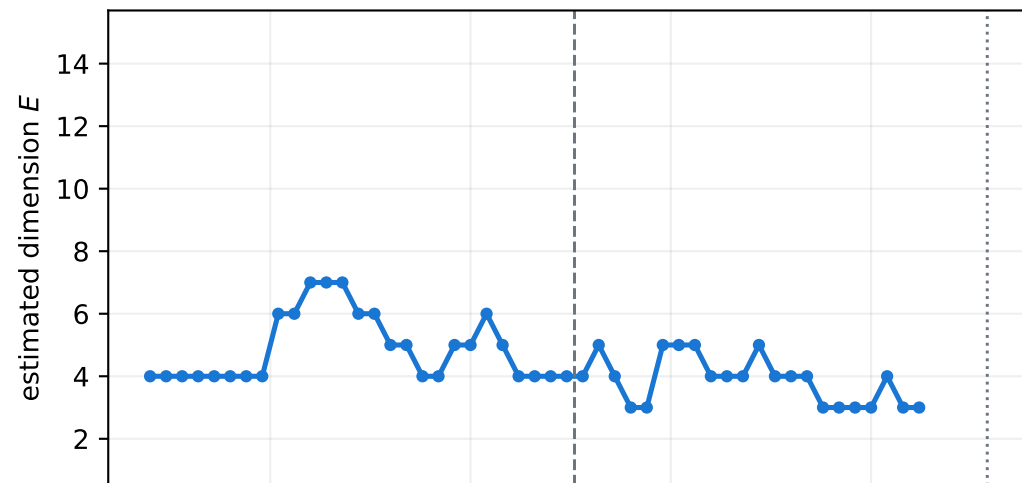
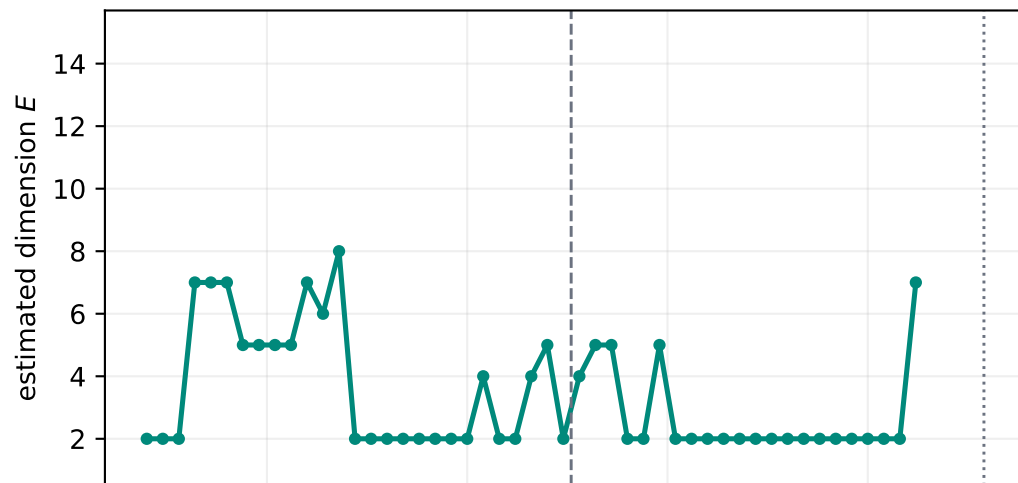


SignMuon: all four methods, fixed $\tau = 1$ (raw loss)

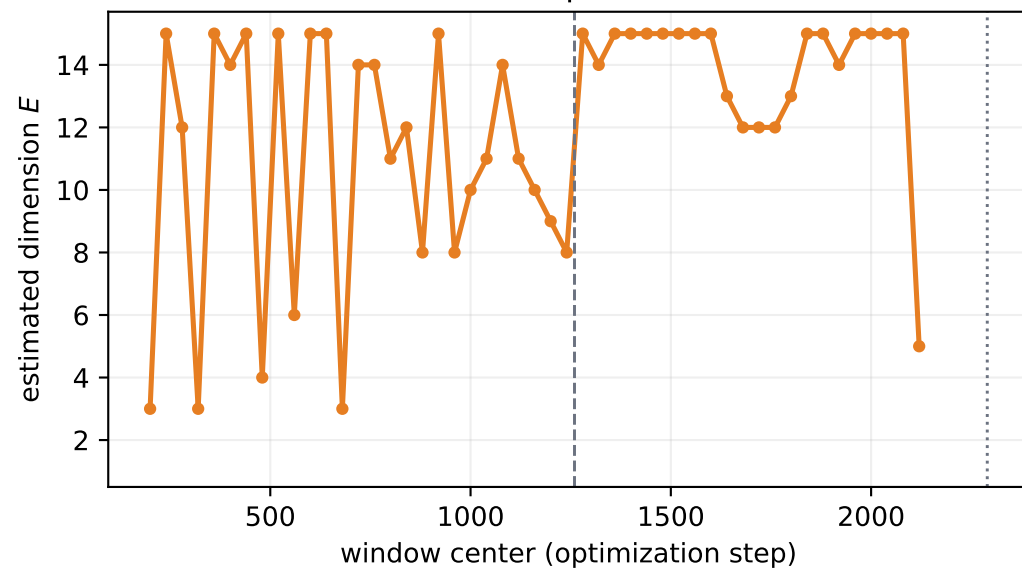
FNN



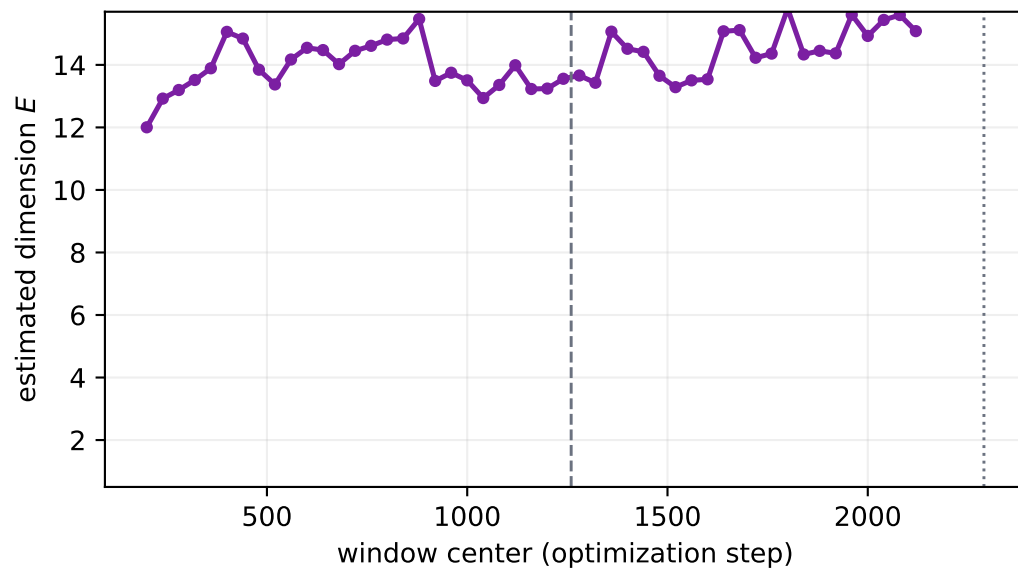
Cao



Simplex



MLE



dashed: start of LR cooldown; dotted: extension start